DSCI 554 Lab 1

Simpson's Paradox, Multiple Testing, and A/B Testing

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Lab Overview

This lab focuses on the Simpson's paradox, multiple testing via two procedures (Bonferroni and false discovery rate corrections), and A/B testing. We will apply most of these concepts to real-life studies.

Lab Mechanics

rubric={mechanics:5}

- Paste the URL to your GitHub repo here: **INSERT YOUR GITHUB REPO URL HERE**
- Once you finish the assignment, you must **knit** this `R` markdown to create a `.pdf` file and push everything to your GitHub repo using `git push`. You are responsible for ensuring all the figures, texts, and equations in the `.pdf` file are appropriately rendered.
- **You must submit the rendered `.pdf` file to Gradescope.**

  Heads-up: You need to have a minimum of 3 commits.

Code Quality

rubric={quality:3}

The code that you write for this assignment will be given one overall grade for code quality. Check our **code quality rubric** as a guide to what we are looking for. Also, for this course (and other MDS courses that use `R`), we are trying to follow the `tidyverse` code style. There is a guide you can refer too: http://style.tidyverse.org/

Each code question will also be assessed for code accuracy (i.e., does it do what it is supposed to do?).

Writing

rubric={writing:3}

To get the marks for this writing component, you should:

- Use proper English, spelling, and grammar throughout your submission (the non-coding parts).
- Be succinct. **This means being specific about what you want to communicate, without being superfluous.**

Check our **writing rubric** as a guide to what we are looking for.

A Note on Challenging Questions

Each lab will have a few challenging questions. These are usually low-risk questions and will contribute to maximum 5% of the lab grade. The main purpose here is to challenge yourself or dig deeper in a particular area. When you start working on labs, attempt all other questions before moving to these questions. If you are running out of time, please skip these questions.

We will be more strict with the marking of these questions. If you want to get full points in these questions, your answers need to

- be thorough, thoughtful, and well-written,
- provide convincing justification and appropriate evidence for the claims you make, and
- impress the reader of your lab with your understanding of the material, your analytical and critical reasoning skills, and your ability to think on your own.

Setup

If you fail to load any packages, you can install them and try loading the library again.

```r
library(datasets)
library(tidyverse)
library(broom)
library(binom)
library(scales)
```

Exercise 1: Simpson's Paradox

The dataset `UCBAdmissions` contains aggregate data from 1973, in the form of six contingency tables, on applicants to graduate school at the University of California, Berkeley (UCB) for the six largest departments (labelled as `A`, `B`, `C`, `D`, `E`, and `F`). Each table shows applicants' absolute frequencies by `Admit` status (`Admitted` or `Rejected`) and `Gender` (`Male` or `Female`).

```r
data(UCBAdmissions)
UCBAdmissions
```

```
## , , Dept = A
##
##           Gender
## Admit      Male Female
##   Admitted  512     89
##   Rejected  313     19
##
## , , Dept = B
##
##           Gender
## Admit      Male Female
##   Admitted  353     17
##   Rejected  207      8
##
## , , Dept = C
##
##           Gender
## Admit      Male Female
##   Admitted  120    202
##   Rejected  205    391
##
## , , Dept = D
##
##           Gender
## Admit      Male Female
##   Admitted  138    131
##   Rejected  279    244
##
## , , Dept = E
##
##           Gender
## Admit      Male Female
##   Admitted   53     94
##   Rejected  138    299
##
## , , Dept = F
##
##           Gender
## Admit      Male Female
##   Admitted   22     24
##   Rejected  351    317
```

Bickel et al. (1975) used this dataset to illustrate Simpson's paradox, as we will see further.

Q1.1. Data Wrangling

rubric={accuracy:4}

Convert the six contingency tables into a single data frame with the following columns: `Admit` (**two-level nominal factor**, `Admitted` or `Rejected`), `Gender` (**two-level nominal factor**, `Male` or `Female`), `Dept` (**six-level nominal factor**; `A`, `B`, `C`, `D`, `E`, or `F`).

The final data frame `UCBAdmissions_training` must have 4526 individual observations**. Make sure that `Rejected`, `Female` and `A` are the baselines.**

```
UCBAdmissions_training <- as_tibble(UCBAdmissions)

# BEGIN SOLUTION
UCBAdmissions_training <- as_tibble(UCBAdmissions)
UCBAdmissions_training <- data.frame(
  Admit = rep(UCBAdmissions_training$Admit, UCBAdmissions_training$n),
  Gender = rep(UCBAdmissions_training$Gender, UCBAdmissions_training$n),
  Dept = rep(UCBAdmissions_training$Dept, UCBAdmissions_training$n)
)

UCBAdmissions_training$Admit <- as.factor(UCBAdmissions_training$Admit)
UCBAdmissions_training$Admit <- fct_relevel(
  UCBAdmissions_training$Admit,
  c("Rejected", "Admitted")
)

UCBAdmissions_training$Gender <- as.factor(UCBAdmissions_training$Gender)
UCBAdmissions_training$Gender <- fct_relevel(
  UCBAdmissions_training$Gender,
  c("Female", "Male")
)

UCBAdmissions_training$Dept <- as.factor(UCBAdmissions_training$Dept)
# END SOLUTION

head(UCBAdmissions_training)
```

```
##      Admit Gender Dept
## 1 Admitted   Male    A
## 2 Admitted   Male    A
## 3 Admitted   Male    A
## 4 Admitted   Male    A
## 5 Admitted   Male    A
## 6 Admitted   Male    A
```

```
tail(UCBAdmissions_training)
```

```
##         Admit Gender Dept
## 4521 Rejected Female    F
## 4522 Rejected Female    F
## 4523 Rejected Female    F
## 4524 Rejected Female    F
## 4525 Rejected Female    F
## 4526 Rejected Female    F
```

Q1.2. First Exploratory Data Analysis

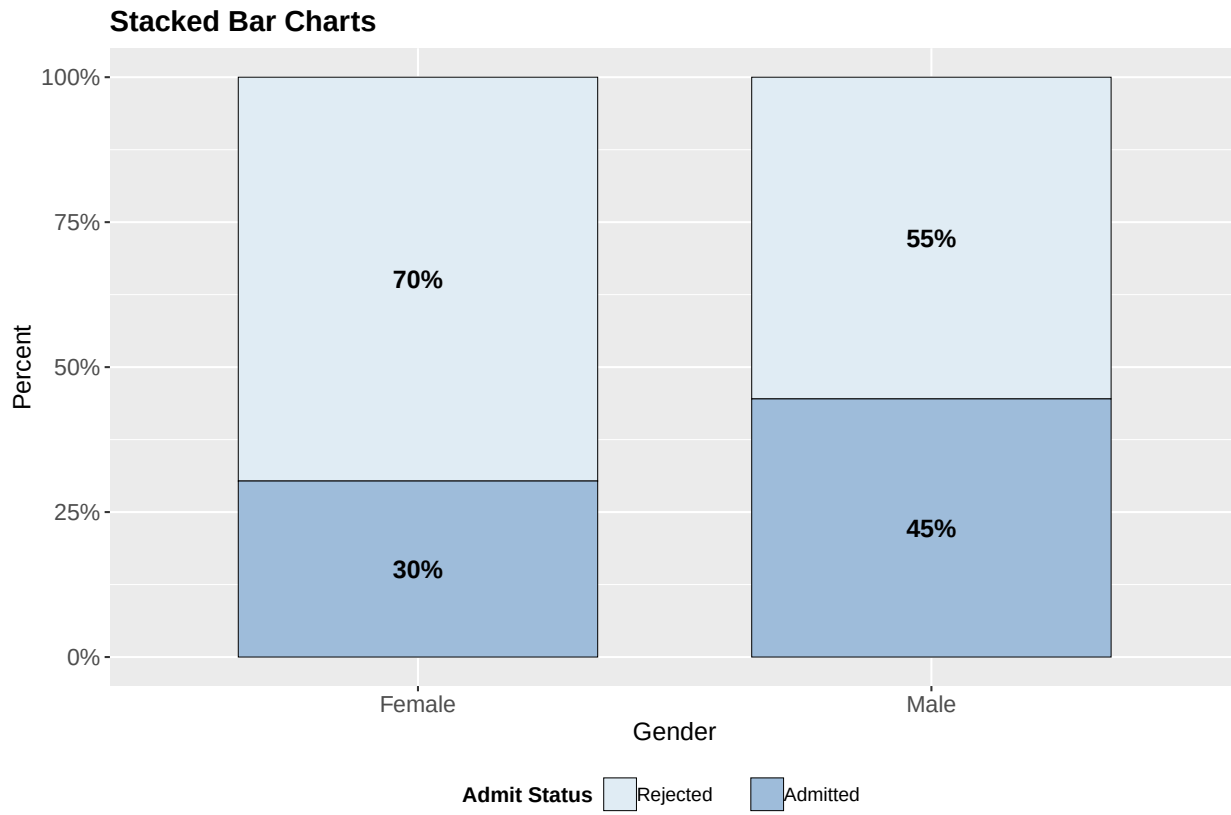
rubric={accuracy:2,viz:4,reasoning:2}

Make a suitable **overall plot** of `Admit` versus `Gender`. **Recall that both variables are categorical.**

```
# BEGIN SOLUTION
UCBAdmissions.prop.summary <- UCBAdmissions_training %>%
  group_by(Gender, Admit) %>%
  summarize(n = length(Gender)) %>%
  mutate(prop = n / sum(n))
```

```
## `summarise()` has grouped output by 'Gender'. You can override using the
## `.groups` argument.
```

```
UCBAdmissions.stacked.bars <- ggplot(UCBAdmissions.prop.summary, aes(
  x = Gender,
  y = prop, fill = Admit
)) +
  geom_bar(stat = "identity", width = 0.7, colour = "black", lwd = 0.1) +
  geom_text(
    aes(label = ifelse(prop >= 0.01,
      paste0(sprintf("%.0f", prop * 100), "%"), ""
    )),
    position = position_stack(vjust = 0.5), colour = "black", fontface = "bold", size = 4.5
  ) +
  scale_y_continuous(labels = percent_format()) +
  labs(y = "Percent", x = "Gender", fill = "") +
  ggtitle("Stacked Bar Charts") +
  theme(
    plot.title = element_text(size = 15, face = "bold"),
    axis.text.x = element_text(size = 12, angle = 0),
    axis.text.y = element_text(size = 12, angle = 0),
    axis.title = element_text(size = 13),
    legend.text = element_text(size = 10, margin = margin(r = 0.5, unit = "cm")),
    legend.title = element_text(size = 11, face = "bold"),
    legend.position = "bottom"
  ) +
  guides(fill = guide_legend(title = "Admit Status")) +
  scale_fill_brewer(palette = "BuPu")

UCBAdmissions.stacked.bars
```




```
# END SOLUTION
```

Graphically speaking, what can we say about the relationship between `Admit` and `Gender`? Is there graphical evidence of sex bias in admission practices during 1973? **Explain in one or two sentences.**

ANSWER:

> *The plot above shows that an overall higher proportion of men were accepted compared to women. This gives us graphical evidence of sex bias in admission practices during 1973.*

Q1.3. First Regression Model

rubric={accuracy:2,reasoning:3}

Using `Admit` as response and `Gender` as explanatory variable, estimate an appropriate regression model called `UCBAdmissions_first_model`. Using $\alpha = 0.05$, interpret the significant regression coefficients. Is there statistical evidence of sex bias in admission practices during 1973? **Explain in two or three sentences.**

ANSWER:

> *By estimating a binomial logistic regression model, we can see that `Gender` is significant for the response `Admit`. Moreover, the estimated regression coefficient indicates that men are 1.84 times more likely to be admitted rather than women. This gives us statistical evidence of sex bias in admission practices during 1973.*

```
UCBAdmissions_first_model <- NULL

# BEGIN SOLUTION
UCBAdmissions_first_model <- glm(Admit ~ Gender,
  family = "binomial",
  data = UCBAdmissions_training
```

```r
)
tidy(UCBAdmissions_first_model,
  conf.int = TRUE,
  exponentiate = TRUE
) %>%
  mutate_if(is.numeric, round, 2)
```

```
## # A tibble: 2 x 7
##   term        estimate std.error statistic p.value conf.low conf.high
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>    <dbl>     <dbl>
## 1 (Intercept)     0.44      0.05    -16.4        0     0.39      0.48
## 2 GenderMale      1.84      0.06      9.55       0     1.62      2.09
```

```r
# END SOLUTION

UCBAdmissions_first_model
```

```
##
## Call:  glm(formula = Admit ~ Gender, family = "binomial", data = UCBAdmissions_training)
##
## Coefficients:
## (Intercept)   GenderMale
##     -0.8305       0.6104
##
## Degrees of Freedom: 4525 Total (i.e. Null);  4524 Residual
## Null Deviance:       6044
## Residual Deviance: 5951  AIC: 5955
```

Q1.4. Second Exploratory Data Analysis

rubric={accuracy:2,viz:4,reasoning:2}

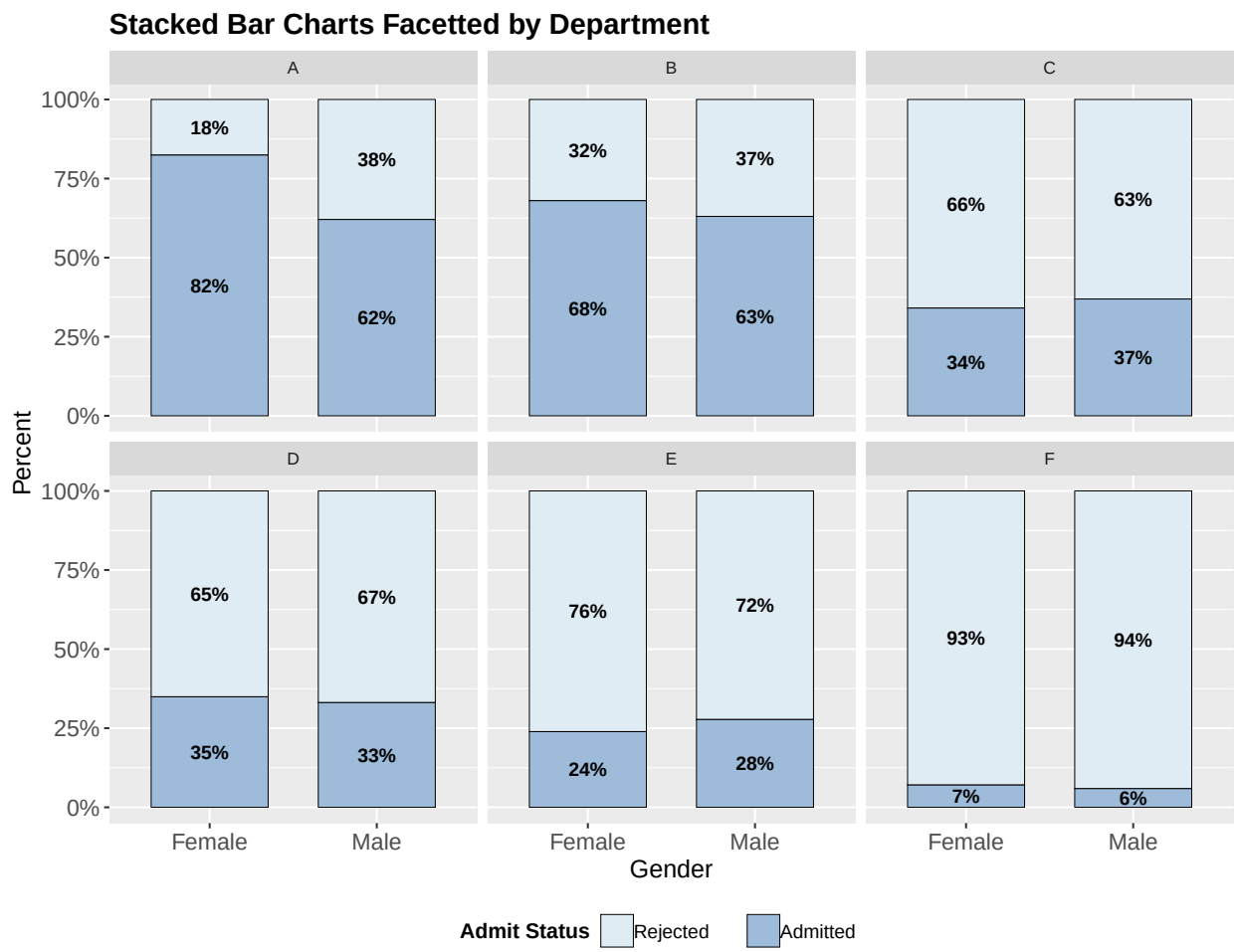
Now, let us control for a second explanatory variable: `Dept`. Use the same plotting arrangement as in **Q1.2** but facetting by `Dept`.

```r
# BEGIN SOLUTION
UCBAdmissions.prop.summary.Dept <- UCBAdmissions_training %>%
  group_by(Dept, Gender, Admit) %>%
  summarize(n = length(Dept)) %>%
  mutate(prop = n / sum(n))
```

```
## `summarise()` has grouped output by 'Dept', 'Gender'. You can override using
## the `.groups` argument.
```

```r
UCBAdmissions.stacked.bars.Dept <- ggplot(UCBAdmissions.prop.summary.Dept, aes(
  x = Gender,
  y = prop, fill = Admit
)) +
  geom_bar(stat = "identity", width = 0.7, colour = "black", lwd = 0.1) +
  geom_text(
    aes(label = ifelse(prop >= 0.03,
      paste0(sprintf("%.0f", prop * 100), "%"), ""
    )),
    position = position_stack(vjust = 0.5), colour = "black", fontface = "bold", size = 3.5
  ) +
  scale_y_continuous(labels = percent_format()) +
```

```r
  labs(y = "Percent", x = "Gender", fill = "") +
  ggtitle("Stacked Bar Charts Facetted by Department") +
  theme(
    plot.title = element_text(size = 15, face = "bold"),
    axis.text.x = element_text(size = 12, angle = 0),
    axis.text.y = element_text(size = 12, angle = 0),
    axis.title = element_text(size = 13),
    legend.text = element_text(size = 10, margin = margin(r = 0.5, unit = "cm")),
    legend.title = element_text(size = 11, face = "bold"),
    legend.position = "bottom"
  ) +
  guides(fill = guide_legend(title = "Admit Status")) +
  facet_wrap(. ~ Dept) +
  scale_fill_brewer(palette = "BuPu")

UCBAdmissions.stacked.bars.Dept
```




```r
# END SOLUTION
```

Graphically speaking, what can we say about the relationship between `Admit` and `Gender` while controlling by `Dept`? Is there graphical evidence of sex bias in admission practices during 1973? **Explain in one or two sentences.**

ANSWER:

The plot above graphically shows no sex bias in admission practices across all departments, except for `A`.

Q1.5. Second Regression Model

rubric={autograde:2,reasoning:2}

Using `Admit` as response with `Gender` and `Dept` as standalone explanatory variables, estimate an appropriate regression model called `UCBAdmissions_second_model`. Using $\alpha = 0.05$, is there statistical evidence of sex bias in admission practices during 1973? **Explain in two or three sentences.**

ANSWER:

By estimating a binomial logistic regression model, we can see that `Gender` is not significant for the response `Admit`. This gives us statistical evidence of no sex bias in admission practices during 1973 when we control by `Dept`.

```r
UCBAdmissions_second_model <- NULL

# BEGIN SOLUTION
UCBAdmissions_second_model <- glm(Admit ~ Dept + Gender,
  family = "binomial",
  data = UCBAdmissions_training
)

tidy(UCBAdmissions_second_model, conf.int = TRUE, exponentiate = TRUE) %>%
  mutate_if(is.numeric, round, 2)
```

```
## # A tibble: 7 x 7
##   term        estimate std.error statistic p.value conf.low conf.high
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>    <dbl>     <dbl>
## 1 (Intercept)     1.98      0.1       6.88    0        1.63      2.4
## 2 DeptB           0.96      0.11     -0.4     0.69     0.77      1.19
## 3 DeptC           0.28      0.11    -11.8     0        0.23      0.35
## 4 DeptD           0.27      0.11    -12.2     0        0.22      0.34
## 5 DeptE           0.18      0.13    -13.8     0        0.14      0.22
## 6 DeptF           0.04      0.17    -19.4     0        0.03      0.05
## 7 GenderMale      0.9       0.08     -1.24    0.22     0.77      1.06
```

```r
# END SOLUTION

UCBAdmissions_second_model
```

```
##
## Call:  glm(formula = Admit ~ Dept + Gender, family = "binomial", data = UCBAdmissions_training)
##
## Coefficients:
## (Intercept)        DeptB        DeptC        DeptD        DeptE        DeptF
##     0.68192     -0.04340     -1.26260     -1.29461     -1.73931     -3.30648
##  GenderMale
##    -0.09987
##
## Degrees of Freedom: 4525 Total (i.e. Null);  4519 Residual
## Null Deviance:       6044
## Residual Deviance: 5187  AIC: 5201
```

(Challenging) Q1.6. Explain the Simpson's Paradox

rubric={reasoning:3}

Explain the Simpson's paradox in this context **in two or three sentences**. What is our overall conclusion?

ANSWER:

> *We can see that Simpson's paradox comes into play here since we first conclude that there is a spurious association between* `Gender` *and* `Admit`*. Nonetheless, when we control for the department (a confounding variable) in our statistical analysis, the previous association is no longer significant. Overall, we can conclude that there is no campus-wide* `Gender` *bias in graduate admissions during 1973.*

Exercise 2: Bonferroni Correction

We will be working with the results from a Genome-wide analysis-like study found in `GWAS_results` from Timbers et al. (2016). The dataset contains two columns: a list of gene names (`gene`) and a list of unadjusted p-values (`pval`) generated from the analysis (the particular statistical test used is the **sequence kernel association test**). These p-values were created by repeating the analysis on many variables from the same dataset. Thus, we have a multiple testing problem to deal with. Each p-value corresponds to a gene and tests whether that gene is associated with a phenotype.

```r
# Load the dataset
GWAS_results <- suppressMessages(read_csv("data/GWAS_results.csv")) %>%
  select(gene = public_gene_name, pval = `p-value`)
head(GWAS_results)
```

```
## # A tibble: 6 x 2
##   gene          pval
##   <chr>        <dbl>
## 1 osm-1   0.00000102
## 2 che-3   0.0000161
## 3 F01D4.9 0.0000556
## 4 mdf-1   0.0001
## 5 cnt-1   0.000124
## 6 lars-1  0.00153
```

Using in a sample of 480 mutant C. elegans (nematode worms), the question in the analysis was: **are there any genes, which when mutated, associated with a phenotype defined as a decrease in the ability to uptake a fluorescent dye into their sensory neurons?** This would indicate there might be a problem with their sensory neurons and that this gene might be important for sensory neuron development or function. The study can be accessed **here**.

Q2.1. Initial Data Insights

rubric={accuracy:2,reasoning:2}

Answer the following questions

- How many genes are present in the total dataset? For this dataset, this corresponds to the number of multiple comparisons that were performed. **Answer in one sentence.**
- How many genes are associated with the phenotype (a decrease in the ability to uptake a fluorescent dye into their sensory neurons) at the unadjusted $\alpha = 0.05$? **Answer in one sentence.**

ANSWER:

> *There are `1150` genes in the dataset, `100` of which are significant at the unadjusted $\alpha = 0.05$.*

Provide the necessary code to support both answers.

```r
# BEGIN SOLUTION
nrow(GWAS_results)
```

```
## [1] 1150
```

```r
nrow(filter(GWAS_results, pval < 0.05))
```

```
## [1] 100
```

```r
# END SOLUTION
```

Q2.2. Multiple Testing Issues

rubric={reasoning:2}

Briefly describe (**in one or two sentences**) why it would be misleading to report only one of the "significant" tests (and ignoring the fact that others were done too).

ANSWER:

> *The problem is p-hacking. The chance of finding at least one falsely significant result (Type I error) when doing multiple tests is much higher than the claimed $\alpha = 0.05$ chance.*

Q2.3. Using `p.adjust()`

rubric={accuracy:1,reasoning:1}

Use the function `p.adjust()` to calculate adjusted p-values via the Bonferroni correction. How many and which genes are associated with the treatment after the adjustment, at the $\alpha = 0.05$ significance level? **Answer in one or two sentences.**

ANSWER:

> *Only two genes, `osm-1` and `che-3`, are statistically associated with the treatment after the adjustment.*

Provide the necessary code to support your answer.

```
# BEGIN SOLUTION
GWAS_results %>%
  mutate(bonf_pval = p.adjust(pval, method = "bonferroni")) %>%
  filter(bonf_pval < 0.05) %>%
  knitr::kable(caption = "Significant findings with Bonferroni correction.")
```

Table 1: Significant findings with Bonferroni correction.

gene	pval	bonf_pval
osm-1	1.00e-06	0.001173
che-3	1.61e-05	0.018515

```
# END SOLUTION
```

Q2.4. Errors in Hypothesis Testing

rubric={reasoning:3}

In a hypothesis test, we limit the Type I error to be no larger than a pre-specified probability, usually $\alpha = 0.05$.

What **probability** does the Bonferroni correction limit? Use the following table to give a concrete definition (the E's and S's are counts of hypothesis testings):

	Null is true	**Alternative is true**
Null rejected	E_1	S_1
Null not rejected	S_2	E_2

Answer in one or two sentences.

ANSWER:

The Bonferroni correction controls the family-wise Type I error rate (FWER) at less than or equal to some pre-specified amount, such as $\alpha = 0.05$*. That is, the probability of making one or more type I error,* $P(E_1 \geq 1)$*, is no more than 0.05.*

Exercise 3: Multiple Comparisons in the Literature

The manuscript "Taste for Competition and the Gender Gap Among Young Business Professionals" (Reuben et al., 2019) analyzes data and made multiple comparisons. The p-values can be found in `competition_results`.

```
competition_results <- suppressMessages(read_csv("data/competition_results.csv"))
head(competition_results)
```

```
## # A tibble: 6 x 2
##   comparison                                p_value
##   <chr>                                       <dbl>
## 1 Fraction choosing the tournament            0
## 2 Probability of ranking 1st in tournament    0.001
## 3 Probability of ranking 1st in piece-rate    0
## 4 Expected rank in tournament                 0
## 5 Risk aversion coefficient                   0.001
## 6 Discount rate                               0.437
```

Q3.1. Study Summary

rubric={reasoning:4}

Write a 3-4 sentence summary of the analysis/study.

> **Hint:** You do **NOT** have to read the entire paper to get this info! For example, you could:
>
> 1. Read the abstract.
> 2. Skim the introduction.
> 3. Skim the results.
> 4. Skim the conclusions.
>
> **Note:** LLMs (such as ChatGPT) can be a valuable tool when used responsibly. Nonetheless, to develop a robust reasoning for future statistical practice both within and outside MDS, LLMs **are not permitted** to answer this question.

ANSWER:

> *Among high-paid business professionals, men earn 9.9% more money than women. This disparity might be accounted for because men enjoy competition more than women do (according to the statistical results in this work,* ***even though it is extremely important to highlight that this may be objected in other works we could find in the literature****). This paper compares 409 male and female business students on 24 measures of taste for competition, finding 12 significant results with no multiple-comparison adjustments.*

Q3.2. Comparing Unadjusted and Adjusted p-values

rubric={accuracy:4}

Using the table from **Q2.4**, the False Discovery Rate (FDR) is defined as

$$\frac{E_1}{E_1 + S_1}.$$

Limiting the FDR to some pre-specified proportion, such as $\delta = 0.05$, controls the proportion of errors among the rejected hypothesis. One FDR-based method is the Benjamini–Hochberg (BH) procedure. Hence, as discussed in class, use this method in addition to the Bonferroni correction.

Take the unadjusted p-values from this analysis (stored in `competition_results`) and perform these two methods to adjust them for multiple comparisons: Bonferroni and FDR corrections.

Then, in a summary table called `table_competition`, count how many tests were statistically significant using the unadjusted p-values, the adjusted p-values from the Bonferroni correction, and the adjusted p-values from the FDR. **Use a significance level of 0.05.**

```
table_competition <- NULL

# BEGIN SOLUTION
table_competition <- competition_results %>%
  select(everything(), pval = p_value) %>%
  mutate(
    bonf_pval = p.adjust(pval, method = "bonferroni"),
    fdr_pval = p.adjust(pval, method = "BH")
  )
tibble(
  Unadjusted = nrow(filter(table_competition, pval < 0.05)),
  Bonferroni = nrow(filter(table_competition, bonf_pval < 0.05)),
  BH = nrow(filter(table_competition, fdr_pval < 0.05))
) %>%
  knitr::kable(caption = "Number of significant findings from each analysis method.")
```

Table 3: Number of significant findings from each analysis method.

Unadjusted	Bonferroni	BH
12	9	12

```
# END SOLUTION

table_competition
```

```
## # A tibble: 24 x 4
##    comparison                                  pval bonf_pval fdr_pval
##    <chr>                                      <dbl>     <dbl>    <dbl>
##  1 Fraction choosing the tournament           0         0      0
##  2 Probability of ranking 1st in tournament 0.001     0.024  0.00267
##  3 Probability of ranking 1st in piece-rate 0         0      0
##  4 Expected rank in tournament                0         0      0
##  5 Risk aversion coefficient                  0.001     0.024  0.00267
##  6 Discount rate                              0.437     1      0.552
##  7 Trust                                      0.088     1      0.151
##  8 Reciprocity                                0.623     1      0.680
##  9 Cooperation                                0.582     1      0.665
## 10 CRT                                        0         0      0
## # i 14 more rows
```

(Challenging) Q3.3. Communicating your Analysis

rubric={reasoning:4}

In no more than 250 words:

- Identify and explain the statistical test(s) used in the part of the analysis where multiple comparisons were made.

- Describe how the authors attempted to control multiple testing issues (or state if they did not make such correction).
- Reflect on how they controlled for multiple testing issues and how the correction choice may have influenced their interpretations of their results (i.e., when we use a different correction method to adjust the p-values, did we get equivalent or very different results?)
- If the analysis did not control multiple comparisons, write a recommendation on what should have been done.

  Note: LLMs (such as ChatGPT) can be a valuable tool when used responsibly. Nonetheless, to develop a robust reasoning for future statistical practice both within and outside MDS, LLMs **are not permitted** to answer this question.

ANSWER:

The p-values were obtained using t-tests for ordinal variables and Chi-squared tests for categorical variables. The authors did not control for multiple comparisons. However, they mentioned in a footnote that the Benjamini-Hochberg procedure does not change the number of significant findings, which we confirmed as true. A Bonferroni correction did change the number of discoveries, though, from 12 to 9. If the authors had used Bonferroni, they would have failed to reject the null hypothesis that there is no difference between genders for the following three measures: Risk aversion coefficient, GPA, and Fraction married.

It seems the authors' interpretations of the results are sound. Whether or not the three measures above are considered discoveries, around half the comparisons yielded tiny p-values, which is enough to suggest that there is indeed a real difference in taste for competitiveness between genders. Some of the experiments may simply be more tightly correlated with that personality trait. Either way, it is not like the authors went hunting for p-values and only came up with one or two that just break the 0.05 threshold. They found fully twelve p-values below 0.05. Since they briefly discussed the Benjamini-Hochberg procedure in a footnote, accurately claiming that the number of discoveries would be the same either way, it seems this study is epistemically honest and well-conceived ***in its statistical findings****.*

Exercise 4: Analysis of an A/B Testing Paper

In this exercise, we will work with the paper "**Improving Library User Experience with A/B Testing: Principles and Process**" by Young (2014). This paper presents a case study where A/B testing is applied with different webpage designs. The primary aim is to compare user interactions to determine which one statistically improves the navigation experience by increasing the homepage click-through rate. The experiment was conducted using the web analytics software Google Analytics and Crazy Egg.

The setup was done on the **Interact** category in the Montana State University's library webpage (more information can be found in the section *Step 1* in the paper). The experimental treatments (as explained and shown in *Step 4* in the paper) are the following: *Interact* (the control treatment), *Connect*, *Learn*, *Help*, and *Services*. The response variable is what we call the click-through rate, i.e., the ratio of users that click on a specific link to the total number of users who view the page (a proportion that goes from 0 to 1).

Firstly, we load the Crazy Egg data from the web.

```r
# Downloading data to a temporary location.
download.file(
  "https://scholarworks.montana.edu/bitstreams/de0a1f89-9c24-4bfe-be4b-289ceb5d7dde/download",
  temp <- tempfile()
)

# Unzipping to a temporary location.
unzip(zipfile = temp, exdir = temp2 <- tempfile())

# Getting the file names of the .csv's and put in a data frame.
crazy_egg <- data.frame(filename = list.files(
  path = temp2, pattern = "*.csv",
  full.names = TRUE,
  recursive = TRUE
))

# Using map to apply read_csv to each filename to read in data.
crazy_egg$raw <- map(as.character(crazy_egg$filename), read_csv)

# Getting rid of temporary files.
unlink(c(temp, temp2))

glimpse(crazy_egg)
```

```
## Rows: 5
## Columns: 2
## $ filename <chr> "/var/folders/hb/j9bdwl_522g4rnvf_fs13nch0000gr/T//RtmpATaHIR~
## $ raw      <list> [<spec_tbl_df[69 x 6]>], [<spec_tbl_df[58 x 6]>], [<spec_tbl~
```

Then, we wrangle data to get the raw data into a tidy data format. This includes making a column that labels which webpage variant (i.e., experimental treatment) it was, as well as unnesting the dataframe so we have a row per observation.

```r
# Creating webpage ID column from filenames.
crazy_egg$webpage <- str_extract(
  string = crazy_egg$filename,
  pattern = "(?<=- )\\w+"
)

# Unnesting data frames to make one big dataframe.
crazy_egg <- unnest(crazy_egg)
```

```r
# Making Column names more readable.
colnames(crazy_egg) <- make.names(colnames(crazy_egg))

glimpse(crazy_egg)
```

```
## Rows: 299
## Columns: 8
## $ filename            <chr> "/var/folders/hb/j9bdwl_522g4rnvf_fs13nch0000gr/T~
## $ Element.ID          <dbl> 128, 69, 61, 67, 78, 98, 62, 118, 50, 87, 76, 96,~
## $ Tag.name            <chr> "area", "a", "input", "a", "a", "a", "button", "a~
## $ Name                <chr> "Montana State University - Home", "FIND", "s.q",~
## $ No..clicks          <dbl> 1291, 842, 508, 166, 151, 102, 101, 55, 46, 42, 3~
## $ Visible.            <lgl> FALSE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, ~
## $ Snapshot.information <chr> "Homepage Version 1 - Interact   •   http://www.l~
## $ webpage             <chr> "Interact", "Interact", "Interact", "Interact", "~
```

Next we make a smaller dataframe that contains just the data we need for calculating click-through rate, which includes:

- Adjusted clicks on homepage (total clicks minus Montana State University - Home clicks).
- Clicks on button target (e.g., Interact, Connect, Learn, Help, Services button).

```r
# Extracting visits.
click_through <- filter(crazy_egg, grepl("created", Snapshot.information)) %>%
  select(webpage, Snapshot.information)

colnames(click_through) <- c("webpage", "clicks")

click_through$clicks <- str_extract(
  string = click_through$clicks,
  pattern = "\\d+(?= clicks)"
) %>%
  as.numeric()

# Extracting homepage clicks.
click_through$home_page_clicks <- filter(crazy_egg, grepl("Home", Name)) %>%
  select(No..clicks) %>%
  unlist()

# Homepage adjusted clicks.
click_through <- mutate(click_through, adjusted_clicks = clicks - home_page_clicks)

# Target clicks (e.g., Interact, Connect, Learn, Help, Services button).
click_through$target_clicks <- filter(crazy_egg, grepl(paste(toupper(click_through$webpage),
  collapse = "|"
), Name)) %>%
  select(No..clicks) %>%
  unlist()

head(click_through)
```

```
## # A tibble: 5 x 5
##   webpage  clicks home_page_clicks adjusted_clicks target_clicks
##   <chr>     <dbl>            <dbl>           <dbl>         <dbl>
## 1 Interact   3714             1291            2423            42
```

```
## 2 Connect    1587              83            1504            53
## 3 Learn      1652              83            1569            21
## 4 Help       1717             122            1595            38
## 5 Services   1348              49            1299            45
```

The `adjusted_clicks` are the **total clicks** we will use to compute the click-through rate by treatment, where `target_clicks` are what we could define as "successes".

Q4.1. Exploratory Data Analysis

rubric={accuracy:3,viz:3,reasoning:3}

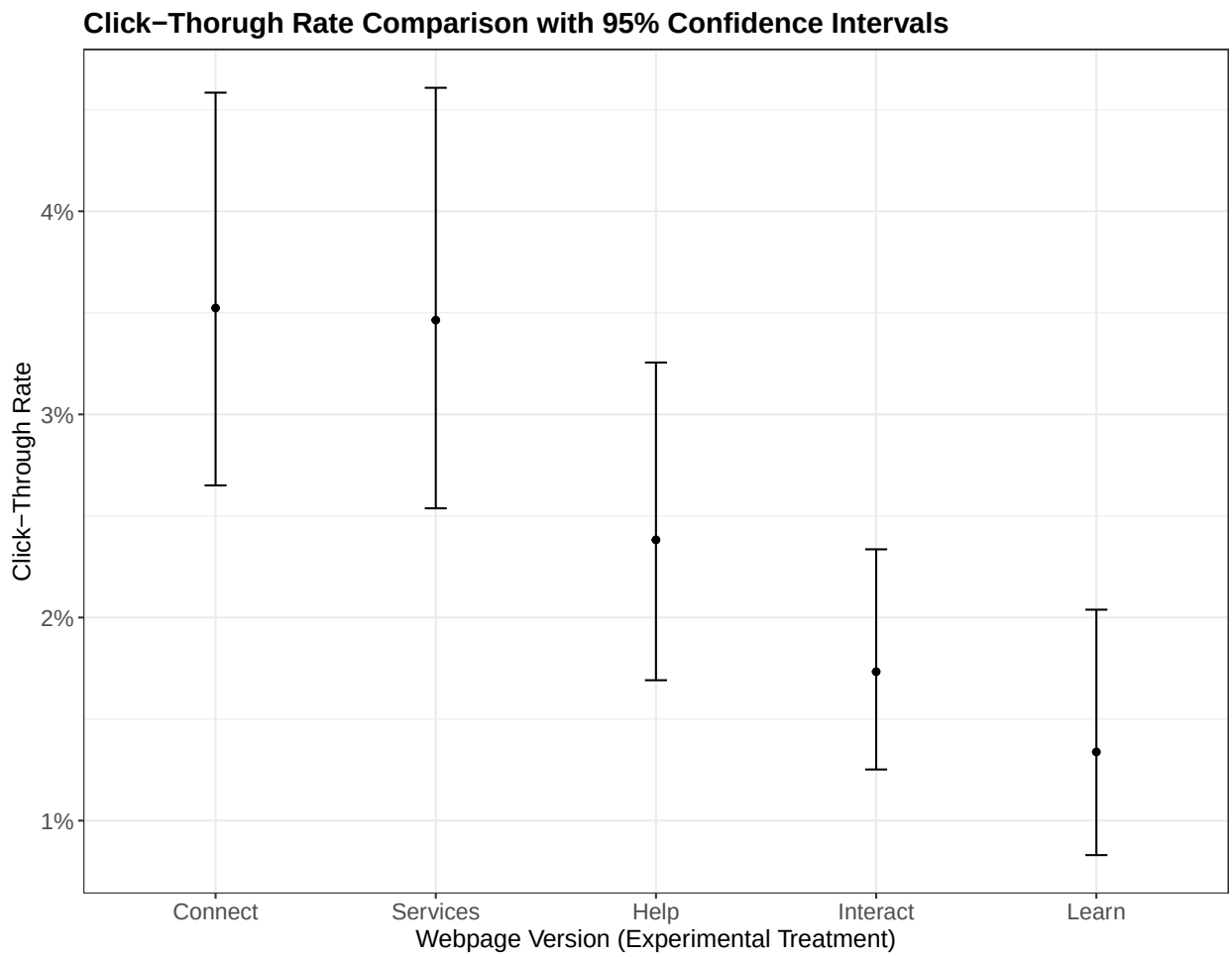
Create a clear and effective visualization of the data, in the form of point estimates, and include confidence intervals (you can use `binom.confint()` from the package `binom`). Firstly, you will have to compute the corresponding click-through rates (call it `click_rate`). Re-order the experimental treatments (i.e., factor levels) in descending order by click-through rate.

```r
# BEGIN SOLUTION
click_through <- mutate(click_through, click_rate = target_clicks / adjusted_clicks)

ci <- binom.confint(click_through$target_clicks,
  click_through$adjusted_clicks,
  methods = "exact"
)
click_through$lower_ci <- ci$lower
click_through$upper_ci <- ci$upper

head(click_through)
```

```
## # A tibble: 5 x 8
##   webpage  clicks home_page_clicks adjusted_clicks target_clicks click_rate
##   <chr>     <dbl>            <dbl>           <dbl>         <dbl>      <dbl>
## 1 Interact   3714             1291            2423            42     0.0173
## 2 Connect    1587               83            1504            53     0.0352
## 3 Learn      1652               83            1569            21     0.0134
## 4 Help       1717              122            1595            38     0.0238
## 5 Services   1348               49            1299            45     0.0346
## # i 2 more variables: lower_ci <dbl>, upper_ci <dbl>
```

```r
# Refactoring webpage version (experimental treatment) based on click-through rate
# in descending order.
click_through$webpage <- fct_reorder(
  click_through$webpage,
  desc(click_through$click_rate)
)

# Plotting as a point with 95% binomial exact confidence intervals.
ggplot(click_through, aes(webpage, click_rate)) +
  geom_point() +
  geom_errorbar(aes(ymin = lower_ci, ymax = upper_ci), width = 0.1) +
  scale_y_continuous(labels = scales::percent) +
  xlab("Webpage Version (Experimental Treatment)") +
  ylab("Click-Through Rate") +
  theme_bw() +
  theme(
    plot.title = element_text(size = 15, face = "bold"),
```

```r
    axis.text.x = element_text(size = 12, angle = 0),
    axis.text.y = element_text(size = 12, angle = 0),
    axis.title = element_text(size = 13),
    legend.text = element_text(size = 10, margin = margin(r = 0.5, unit = "cm")),
    legend.title = element_text(size = 11, face = "bold"),
    legend.position = "bottom"
  ) +
  ggtitle("Click-Thorugh Rate Comparison with 95% Confidence Intervals")
```




```r
# END SOLUTION
```

What can we conclude from this plot? **In three or four sentences**, point out any interesting finding coming from the confidence intervals.

ANSWER

> *The Connect version of the webpage had the highest click-through rate, followed by Services, Help, Interact (i.e., the control treatment), and finally Learn. We see that only the confidence intervals of Connect and Services do not overlap with the confidence interval of the Interact version of the webpage, indicating that Connect and Services are statistically significant different (and higher) click-through rates compared to the Interact version of the page. The confidence interval of the Learn version of the webpage does not overlap with the confidence intervals of Services and Connect, indicating that the Learn version of the webpage is statistically significant different (and lower) than these other two pages in terms of the click-through rate.*

Q4.2. Pairwise Comparisons

rubric={accuracy:2,reasoning:2}

Perform **pairwise** frequentist hypothesis test analyses to assess whether there are statistically significant differences between all the experimental treatments (use $\alpha = 0.05$). **Do not forget to control for multiple comparisons** (you can use either one of the methods seen throughout this lab). Ensure you are assuming a reasonable distribution for the data when choosing the statistical analysis for your hypothesis test.

Does your analysis agree or disagree with the insight from your plot?

ANSWER:

> *We use the hypothesis test for two population proportions, which uses a normal approximation. Both correction methods show that there are significant differences in the following pairs:*
>
> - *Connect and Interact.*
> - *Connect and Learn.*
> - *Services and Interact.*
> - *Services and Learn.*

```r
# BEGIN SOLUTION
successes <- click_through$target_clicks
trials <- click_through$adjusted_clicks
names(successes) <- click_through$webpage
names(trials) <- click_through$webpage

pairwise.prop.test(successes,
  trials,
  p.adjust.method = "bonferroni"
)
```

```
##
##  Pairwise comparisons using Pairwise comparison of proportions
##
## data:  successes out of trials
##
##          Interact Connect Learn  Help
## Connect  0.0057   -       -      -
## Learn    1.0000   0.0013  -      -
## Help     1.0000   0.7594  0.4144 -
## Services 0.0129   1.0000  0.0026 1.0000
##
## P value adjustment method: bonferroni
```

```r
pairwise.prop.test(successes,
  trials,
  p.adjust.method = "BH"
)
```

```
##
##  Pairwise comparisons using Pairwise comparison of proportions
##
## data:  successes out of trials
##
##          Interact Connect Learn  Help
## Connect  0.0019   -       -      -
## Learn    0.4405   0.0013  -      -
## Help     0.2312   0.1266  0.0829 -
```

```
## Services 0.0032   1.0000  0.0013 0.1497
##
## P value adjustment method: BH
```

```
# END SOLUTION
```

Submission

Congratulations! You are done the lab. Do not forget to:

- Knit the assignment to generate the `.pdf` file and push everything to your Github repo.
- Double check all the figures, texts, equations are rendered properly in the `.pdf` file
- **Submit the `.pdf` file to Gradescope.**

References

- Bickel, P., Hammel, E., and O'Connell, J. (1975). Sex Bias in Graduate Admissions: Data from Berkeley. Science, 187(4175), 398-404. Retrieved February 17, 2021, from http://www.jstor.org/stable/1739581-
- Reuben, E., Sapienza, P., and Zingales, L. (2019). Taste for Competition and the Gender Gap among Young Business Professionals, Working Papers 20190031, New York University Abu Dhabi, Department of Social Science, revised Nov 2019.
- Timbers, T., Garland S., Mohan S., Flibotte S., Edgley M, et al. (2016). Accelerating Gene Discovery by Phenotyping Whole-Genome Sequenced Multi-mutation Strains and Using the Sequence Kernel Association Test (SKAT). PLOS Genetics 12(8): e1006235. https://doi.org/10.1371/journal.pgen.1006235
- Young, S. (2014). Improving Library User Experience with A/B Testing: Principles and Process. Weave: Journal of Library User Experience. 1. 10.3998/weave.12535642.0001.101.